Nokia BLR-CORE-02 Core Node Standard Operating Procedure for Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding)

Node: BLR-CORE-02 | **Vendor:** Nokia | **Severity:** Critical | **Alarm Code:** ALARM_2432

Description

This Standard Operating Procedure (SOP) is designed to address the critical failure domain of Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on the Nokia BLR-CORE-02 Core Node. This failure domain occurs due to the underlying mechanics of the Routing Protocol Flap, which is characterized by a rapid oscillation between the establishment and teardown of routing protocol sessions. This phenomenon is often caused by a combination of factors, including, but not limited to, misconfigured routing protocols, hardware issues, or software bugs. In the context of the Nokia BLR-CORE-02 Core Node, the Routing Protocol Flap can be triggered by the interaction between the BGP, OSPF, and ISIS protocols, which are configured to operate on the node. The protocol flap can result in the loss of routing protocol adjacencies, leading to suboptimal routing and potential network instability. It is essential to identify and address the root cause of the protocol flap to prevent network downtime and ensure the continued operation of the BLR-CORE-02 Core Node.

Pre-requisite Checks

- Verify that the BLR-CORE-02 Core Node is reachable and accessible for remote troubleshooting.

- Check the status of the redundant links connected to the BLR-CORE-02 Core Node to ensure that they are operational and configured correctly.

- Review the system logs for any error messages or warnings related to the routing protocols.

- Confirm that the routing protocol configurations are consistent across all nodes in the network.

Potential Root Causes

Cause	Probability
Misconfigured routing protocols	High
Hardware issues	Medium
Software bugs	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP summary information, Description: This command displays a summary of the BGP routing protocol, including the number of established and active connections, the number of prefixes advertised, and the number of prefixes received. This information can help identify any issues related to BGP adjacency loss or prefix flooding.
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```
Command: show ospf neighbor, Expected Output: OSPF neighbor information, Description: This command displays information about the OSPF neighbors, including the neighbor's IP address, the neighbor's state (UP/UP/DOWN/DOWN), and the number of LSA advertisements received. This information can help identify any issues related to OSPF adjacency loss or LSA flooding.

Command: show isis lsp flooding, Expected Output: ISIS LSP flooding information, Description: This command displays information about the ISIS LSP flooding, including the number of LSPs flooded, the number of LSPs received, and the number of LSPs rejected. This information can help identify any issues related to ISIS LSP flooding or routing protocol flap.
```

Resolution Steps

Step 1: Step: Verify the routing protocol configuration, Command: show bgp configuration, Description: Verify that the BGP configuration is correct and consistent across all nodes in the network. Check for any misconfigured routing policies, neighbor addresses, or update groups.

Step 2: Step: Check for hardware issues, Command: show hardware, Description: Check the hardware status of the BLR-CORE-02 Core Node to ensure that there are no issues related to the routing protocol flap.

Step 3: Step: Identify and fix software bugs, Command: show software, Description: Identify any software bugs that may be causing the routing protocol flap and apply the necessary software patches or updates.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP summary information, Description: Verify that the BGP routing protocol is operating correctly and that there are no issues related to BGP adjacency loss or prefix flooding.

Command: show ospf neighbor, Expected Output: OSPF neighbor information, Description: Verify that the OSPF neighbors are operating correctly and that there are no issues related to OSPF adjacency loss or LSA flooding.

Command: show isis lsp flooding, Expected Output: ISIS LSP flooding information, Description: Verify that the ISIS LSP flooding is operating correctly and that there are no issues related to ISIS LSP flooding or routing protocol flap.
```

Roll-back Procedures

- Step: Restore the previous configuration, Command: copy running-config startup-config, Description: Restore the previous configuration to ensure that the network returns to its previous state.

- Step: Roll back to the previous software version, Command: software rollback, Description: Roll back to the previous software version to ensure that the network returns to its previous state.